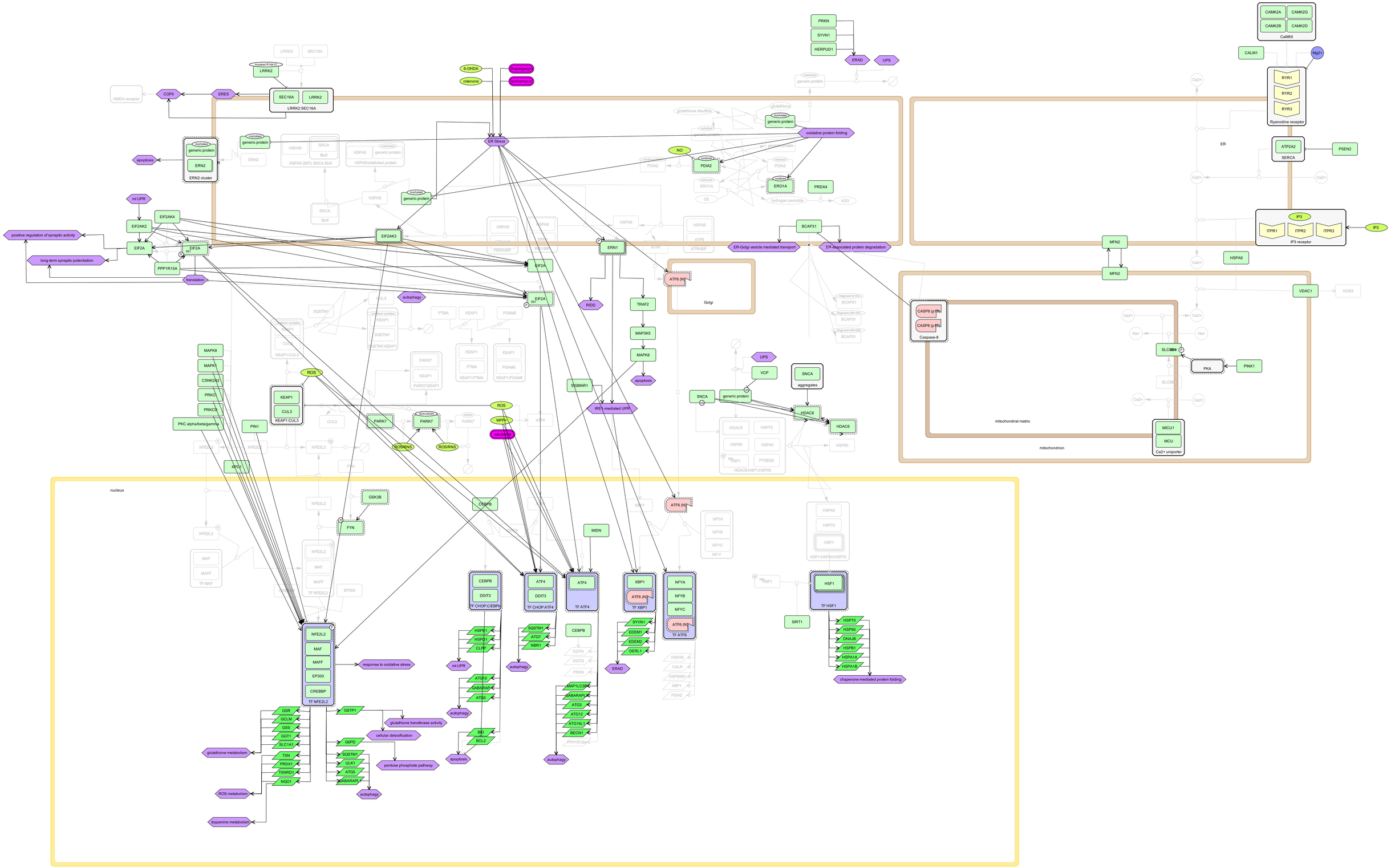

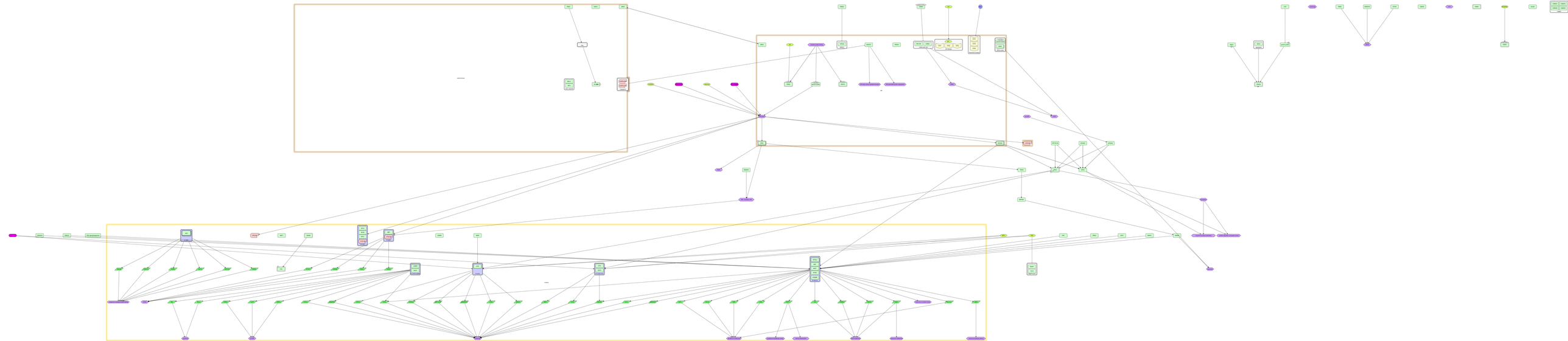

mutated R1441C
LRRK2
SEC16A
LRRK2
LRRK2:SEC16A
ERES
COPII
NMDA receptor
6-OHDA
rotenone
thapsigargin
tunicamycin
ER Stress
PRKN
SYVN1
HERPUD1
ERAD
UPS
misfolded
generic protein
misfolded
generic protein
ERN2
ERN2 cluster
apoptosis
ERN2
HSPA5
SNCA
fibril
HSPA5 (BIP): SNCA fibril
HSPA5
generic protein
HSPA5:misfolded protein
HSPA5
SNCA
fibril
misfolded
generic protein
glutathione disulfide
glutathione
misfolded
generic protein
oxidative protein folding
unfolded
generic protein
generic protein
NO
S-nitrosylation
PDIA2
oxidized
PDIA2
reduced
PDIA2
reduced
ERO1A
oxidized
ERO1A
PRDX4
O2
hydrogen peroxide
H2O
mt UPR
EIF2AK4
EIF2AK2
EIF2A
EIF2A
S51
P
PPP1R15A
translation
positive regulation of synaptic activity
long-term synaptic potentiation
EIF2AK3
HSPA5
EIF2AK3
PERK:BIP
HSPA5
ERN1
IRE1:BIP
HSPA5
ATF6
HSPA5
ATF6
ATF6:BIP
P
ERN1
EIF2A
EIF2A
S51
RIDD
TRAF2
MAP3K5
MAPK8
apoptosis
ATF6 (N)
Golgi
BCAP31
ER-Golgi vesicle mediated transport
ER-associated protein degradation
fragment 2-164
BCAP31
fragment 235-167
BCAP31
fragment 238-246
BCAP31
CUL3
autophagy
SQSTM1
oxidized cysteine
KEAP1
CUL3
KEAP1:CUL3
oxidized cysteine
KEAP1
SQSTM1
SQSTM1:KEAP1
PTMA
KEAP1
PGAM5
KEAP1
PTMA
KEAP1:PTMA
KEAP1
PGAM5
KEAP1:PGAM5
PARK7
KEAP1
PARK7:KEAP1
MAPK8
MAPK1
CSNK2A2
PRKCI
PRKCD
PKC alpha/beta/gamma
PIN1
ROS
KEAP1
CUL3
KEAP1:CUL3
CUL3
PARK7
SOH/SO2H
PARK7
SO3H
PARK7
ROS
MPP+
salubrinal
ATF4
ROS/RNS
ROS/RNS
NFE2L2
NFE2L2
NFE2L2
Ub
XPO1
FYN
SIGMAR1
IRE1-mediated UPR
UPS
VCP
Ub
generic protein
SNCA
Ub
SNCA
aggregates
HDAC6
HDAC6
HDAC6
HSP70
HSP90
HSP40
Ac
K80
HSF1
PTGES3
HDAC6:HSF1:HSP90
HSP90
nucleus
NFE2L2
GSK3B
P
FYN
CEBPB
DDIT3
ATF4
XBP1
ATF6 (N)
NFYA
NFYB
NFYC
NF-Y
HSP40
HSP70
HSF1
HSF1:HSP40:HSP70
NFE2L2
R
MAF
MAFF
TF MAF
NFE2L2
MAF
MAFF
TF NFE2L2
CREBBP
EP300
MIDN
CEBPB
DDIT3
TF CHOP:C/EBPb
ATF4
DDIT3
TF CHOP:ATF4
ATF4
TF ATF4
XBP1
ATF6 (N)
TF XBP1
NFYA
NFYB
NFYC
ATF6 (N)
TF ATF6
Ac
K80
HSF1
HSF1
TF HSF1
SIRT1
NFE2L2
P
MAF
MAFF
EP300
CREBBP
TF NFE2L2
response to oxidative stress
HSPE1
HSPD1
CLPP
mt UPR
SQSTM1
ATG7
NBR1
autophagy
CEBPB
DDIT4
DDIT3
PRKN
SYVN1
EDEM1
EDEM2
DERL1
ERAD
HSPA5
CALR
HSP90B1
XBP1
PDIA2
HSP70
HSP90
DNAJB
HSPB1
HSPA1A
HSPA1B
chaperone-mediated protein folding
ATG10
GABARAP
ATG5
autophagy
MAP1LC3B
GABARAPL2
ATG3
ATG12
ATG16L1
BECN1
PPP1R15A
BID
BCL2
apoptosis
autophagy
GSR
GCLM
GSS
GGT1
SLC1A1
glutathione metabolism
TXN
PRDX1
TXNRD1
NQO1
ROS metabolism
dopamine metabolism
GSTP1
glutathione transferase activity
cellular detoxification
G6PD
pentose phosphate pathway
SQSTM1
ULK1
ATG5
GABARAPL1
autophagy
CaMKII
CAMK2A
CAMK2G
CAMK2B
CAMK2D
CALM1
Mg2+
RYR1
RYR2
RYR3
Ryanodine receptor
Ca2+
ER
ATP2A2
SERCA
PSEN2
Ca2+
Ca2+
IP3
ITPR1
ITPR2
ITPR3
IP3 receptor
IP3
MFN2
MFN2
HSPA9
Ca2+
VDAC1
NOS3
CASP8 (p18)
CASP8 (p10)
Caspase-8
Ca2+
Ca2+
Na+
Na+
SLC8B1
P
PKA
PINK1
SLC8B1
Ca2+
Ca2+
mitochondrial matrix
mitochondrion
MICU1
MCU
Ca2+ uniporter

mitochondrion
PINK1
VDAC1
MFN2
PKA
MICU1
MCU
Ca2+ uniporter
SLC8B1
P
CASP8 (p18)
CASP8 (p10)
Caspase-8
6-OHDA
tunicamycin
rotenone
thapsigargin
PSEN2
LRRK2
IP3
Mg2+
MFN2
NO
oxidative protein folding
ATP2A2
SERCA
BCAP31
PRDX4
SEC16A
LRRK2
LRRK2:SEC16A
IP3
ITPR1
ITPR2
ITPR3
IP3 receptor
RYR1
RYR2
RYR3
Ryanodine receptor
generic protein
ERN2
ERN2 cluster
PDIA2
generic protein
ERO1A
ER-Golgi vesicle mediated transport
ER-associated protein degradation
ER
ERES
ER Stress
mt UPR
COPII
ERN1
EIF2AK3
ATF6 (N)
RIDD
SIGMAR1
IRE1-mediated UPR
TRAF2
MAP3K5
PPP1R15A
EIF2AK4
EIF2AK2
EIF2A
EIF2A
translation
long-term synaptic potentiation
positive regulation of synaptic activity
apoptosis
SNCA
SNCA
aggregates
VCP
generic protein
HDAC6
autophagy
PRKN
HERPUD1
SYVN1
ERAD
HSPA9
UPS
PARK7
ROS/RNS
PARK7
CALM1
CAMK2A
CAMK2G
CAMK2B
CAMK2D
CaMKII
salubrinal
CSNK2A2
PRKCD
PKC alpha/beta/gamma
HSF1
TF HSF1
ATF6 (N)
SIRT1
GSK3B
NFYA
NFYB
NFYC
ATF6 (N)
TF ATF6
XBP1
ATF6 (N)
TF XBP1
CEBPB
MIDN
DNAJB
HSPA1A
HSPB1
HSP70
HSPA1B
HSP90
FYN
SYVN1
EDEM2
DERL1
EDEM1
CEBPB
DDIT3
TF CHOP:C/EBPβ
ATF4
TF ATF4
ATF4
DDIT3
TF CHOP:ATF4
nucleus
NFE2L2
MAF
MAFF
EP300
CREBBP
TF NFE2L2
MPP+
ROS
PIN1
PRKCI
XPO1
MAPK1
MAPK8
KEAP1
CUL3
KEAP1:CUL3
chaperone-mediated protein folding
ERAD
BID
BCL2
HSPE1
CLPP
HSPD1
ATG10
GABARAP
ATG12
ATG5
ATG16L1
MAP1LC3B
GABARAPL2
ATG3
BECN1
NBR1
ATG7
SQSTM1
ULK1
GABARAPL1
GGT1
GCLM
GSR
GSS
GSTP1
TXN
TXNRD1
PRDX1
NQO1
response to oxidative stress
SLC1A1
G6PD
apoptosis
mt UPR
autophagy
glutathione metabolism
glutathione transferase activity
cellular detoxification
ROS metabolism
dopamine metabolism
pentose phosphate pathway